

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0791

Smith 345 kV
August 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0791. The Interconnection Customer proposes a 561.7 MW natural gas combined cycle facility interconnecting at the Smith 345 kV substation in the Pennsylvania area. The request was received on May 04, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 334 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$118,781,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: DOS AMIGOS – NEWHALL 138 KV
Contingency: MEDORA – P.T.S 345 KV
Loading: 247.0% (Rating: 190 MVA)
DFAX: 5.9%
Pre-project loading: 229.6%
2. Facility: NEWHALL – CARSON 138 KV L
Contingency: MEDORA – P.T.S 345 KV L
Loading: 226.1% (Rating: 214 MVA)
DFAX: 5.9%
Pre-project loading: 210.7%
3. Facility: CARSON – FORBES 138 KV line
Contingency: P.T.S SUBSTATION – MEDORA 345 KV
Loading: 217.1% (Rating: 230 MVA)
DFAX: 5.9%
Pre-project loading: 202.7%
4. Facility: FORBES – US GYPSUM SUBSTATION 138 KV line
Contingency: P.T.S – MEDORA 345 KV
Loading: 206.7% (Rating: 254 MVA)
DFAX: 5.9%
Pre-project loading: 193.6%
5. Facility: CASCADE VALLEY – VARICK 138 KV
Contingency: SMITH – HEMET 345 KV L
Loading: 176.3% (Rating: 230 MVA)
DFAX: 38.0%
Pre-project loading: 83.4%

6. Facility: VARICK – HIGH FALLS HY 138 KV
Contingency: SMITH – HEMET 345 KV
Loading: 173.4% (Rating: 237 MVA)
DFAX: 38.0%
Pre-project loading: 83.6%
7. Facility: HIGH FALLS HY – SMITH 138 KV line
Contingency: HEMET – SMITH 345 KV
Loading: 160.9% (Rating: 256 MVA)
DFAX: 53.3%
Pre-project loading: 44.1%
8. Facility: US GYPSUM – LEICESTER JUNCTION 138 KV
Contingency: P.T.S – MEDORA 345 KV
Loading: 160.5% (Rating: 241 MVA)
DFAX: 6.0%
Pre-project loading: 146.5%
9. Facility: LEICESTER JCT – WOODVILLE 138 KV Line
Contingency: MEDORA SUBSTATION – P.T.S SUBSTATION 345 KV
Loading: 158.4% (Rating: 258 MVA)
DFAX: 6.0%
Pre-project loading: 145.3%
10. Facility: HEMET – WOODVILLE 138 KV
Contingency: COLLIER – HOYT 138 KV
Loading: 145.8% (Rating: 259 MVA)
DFAX: 9.6%
Pre-project loading: 124.9%
11. Facility: SIERRA BROOKS – CONRATH DPC SOLAR 138 KV
Contingency: US GYPSUM – FORBES 138 KV
Loading: 136.2% (Rating: 272 MVA)
DFAX: 5.6%
Pre-project loading: 124.6%
12. Facility: WALTER L ANDERSON – SIERRA BROOKS 138 KV line
Contingency: FORBES – US GYPSUM 138 KV
Loading: 131.7% (Rating: 266 MVA)
DFAX: 5.6%
Pre-project loading: 119.9%
13. Facility: LEWISVILLE – OAKLAND 138 KV L
Contingency: FORBES – US GYPSUM 138 KV
Loading: 130.5% (Rating: 221 MVA)
DFAX: 5.6%
Pre-project loading: 116.3%
14. Facility: SMITH 345-138 kV Autotransformer
Contingency: SMITH – HEMET 345 KV
Loading: 129.9% (Rating: 424 MVA)
DFAX: 100.0%
Pre-project loading: 2.5%
15. Facility: OAKLAND – WALTER L ANDERSON 138 KV
Contingency: FORBES – US GYPSUM 138 KV Line
Loading: 127.8% (Rating: 244 MVA)
DFAX: 5.6%
Pre-project loading: 115.0%
16. Facility: HAMPTON RD – FILER 345 KV Line
Contingency: FISHKILL PLAINS – DILLARD-ORANGE 345 KV Line

- Loading: 119.6% (Rating: 1,216 MVA)
DFAX: 28.0%
Pre-project loading: 106.7%
17. Facility: JULIA – GROVE POINT RD 138 KV
Contingency: MIDDLE FORK – ULMERTON TRANSFER CTR 345 KV
Loading: 116.2% (Rating: 157 MVA)
DFAX: 6.4%
Pre-project loading: 93.1%
18. Facility: CANYONVILLE – SARGENT CANYON 138 KV
Contingency: SMITH – HEMET 345 KV line
Loading: 115.0% (Rating: 170 MVA)
DFAX: 20.3%
Pre-project loading: 47.9%
19. Facility: PINEY CREEK PROJECT – LIONS RD 500 KV
Contingency: EL PASO – GANSO 500 KV
Loading: 110.8% (Rating: 3,166 MVA)
DFAX: 11.2%
Pre-project loading: 108.8%
20. Facility: N COWAN – MONTOUR 138 KV
Contingency: HEMET – WOODVILLE 138 KV L
Loading: 110.4% (Rating: 176 MVA)
DFAX: 5.8%
Pre-project loading: 91.9%
21. Facility: STEWART MTN DAM – RITTER PARK 345 KV
Contingency: P.T.S – MEDORA 345 KV
Loading: 108.3% (Rating: 1,760 MVA)
DFAX: 11.8%
Pre-project loading: 104.5%
22. Facility: HANDSOME LAKE – PINEY CRK PROJECT 500 KV
Contingency: PINEY CRK PROJECT – LIONS RD 500 KV Line
Loading: 108.1% (Rating: 2,747 MVA)
DFAX: 7.6%
Pre-project loading: 106.5%
23. Facility: HOYT – COLLIER 138 KV
Contingency: WOODVILLE – HEMET 138 KV
Loading: 108.0% (Rating: 264 MVA)
DFAX: 5.8%
Pre-project loading: 95.7%
24. Facility: CASCADE VLY – CANYONVILLE SUBSTATION 138 KV
Contingency: SMITH – HEMET 345 KV
Loading: 107.8% (Rating: 191 MVA)
DFAX: 20.3%
Pre-project loading: 47.9%
25. Facility: ULMERTON TRANSFER CTR – STEWART MOUNTAIN DAM 345 KV L
Contingency: MEDORA – P.T.S 345 KV line
Loading: 106.8% (Rating: 1,238 MVA)
DFAX: 16.1%
Pre-project loading: 99.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEWHALL – DOS AMIGOS 138 KV Line	Rebuild line with higher-capacity conductor	\$1,612,000	100% (\$1,612,000)
CARSON – NEWHALL 138 KV	Rebuild line with higher-capacity conductor	\$967,000	100% (\$967,000)
CARSON – FORBES 138 KV line	Rebuild line with higher-capacity conductor	\$2,113,000	100% (\$2,113,000)
FORBES – US GYPSUM 138 KV	Rebuild line with higher-capacity conductor	\$1,398,000	100% (\$1,398,000)
CASCAE VALLEY – VARIK 138 KV	Rebuild line with higher-capacity conductor	\$4,395,000	100% (\$4,395,000)
VARICK – HIGH FALLS HY 138 KV	Rebuild line with higher-capacity conductor	\$1,393,000	100% (\$1,393,000)
SMITH – HIGH FALLS HY 138 KV	Rebuild line with higher-capacity conductor	\$1,823,000	100% (\$1,823,000)
US GYPSUM – LEICESTER JCT 138 KV	Rebuild line with higher-capacity conductor	\$4,785,000	100% (\$4,785,000)
LEICESTER JCT – WOODVILLE 138 KV Line	Rebuild line with higher-capacity conductor	\$1,387,000	100% (\$1,387,000)
HEMET – WOODVILLE 138 KV	Rebuild line with higher-capacity conductor	\$13,977,000	100% (\$13,977,000)
SIERRA BROOKS – CONRATH DPC SOLAR 138 KV	Rebuild line with higher-capacity conductor	\$780,000	100% (\$780,000)
WALTER L ANDERSON – SIERRA BROOKS 138 KV line	Rebuild line with higher-capacity conductor	\$4,388,000	100% (\$4,388,000)
LEWISVILLE – OAKLAND 138 KV L	Rebuild line with higher-capacity conductor	\$2,644,000	100% (\$2,644,000)
SMITH 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$14,814,000	100% (\$14,814,000)
OAKLAND – WALTER L ANDERSON 138 KV	Rebuild line with higher-capacity conductor	\$1,921,000	100% (\$1,921,000)
HAMPTON RD – FILER 345 KV Line	Rebuild line with higher-capacity conductor	\$57,087,000	38% (\$21,922,000)
JULIA – GROVE POINT RD 138 KV	Rebuild line with higher-capacity conductor	\$1,715,000	100% (\$1,715,000)
CANYONVILLE – SARGENT CANYON 138 KV	Rebuild line with higher-capacity conductor	\$2,802,000	100% (\$2,802,000)
LIONS RD – PINEY CREEK PROJECT 500 KV	Rebuild line with higher-capacity conductor	\$242,961,000	9% (\$22,836,000)
MONTOUR – N COWAN 138 KV	Rebuild line with higher-capacity conductor	\$3,807,000	100% (\$3,807,000)
STEWART MTN DAM – RITTER PARK 345 KV	Reconductor line	\$8,866,000	10% (\$883,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HANDSOME LAKE – PINEY CRK PROJECT 500 KV	Reconductor line	\$39,710,000	8% (\$3,248,000)
HOYT – COLLIER 138 KV	Reconductor line	\$858,000	100% (\$858,000)
CASCADE VLY – CANYONVILLE SUBSTATION 138 KV	Reconductor line	\$931,000	100% (\$931,000)
ULMERTON TRANSFER CTR – STEWART MTN DAM 345 KV	Reconductor line	\$6,005,000	23% (\$1,382,000)
	Total allocated to project		\$118,781,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.